string is not an indication of the magnitude of the force.

8 **D** By the principle of conservation of energy,

Loss in GPE = Work Done by (mean) force on spike (as it enters the ground)

$mg(h+x) = Fx$ (where m is the mass of spike, h is 4.5 m, x is 0.25 m)

$$F = \frac{mg(h+x)}{x} = \frac{(3.0)(9.81)(4.5+0.25)}{0.25} = 560 \text{ N (2 s.f)}$$

9 **B** When the motor is not spinning, the 60 N mass will move downwards as there is a net downward